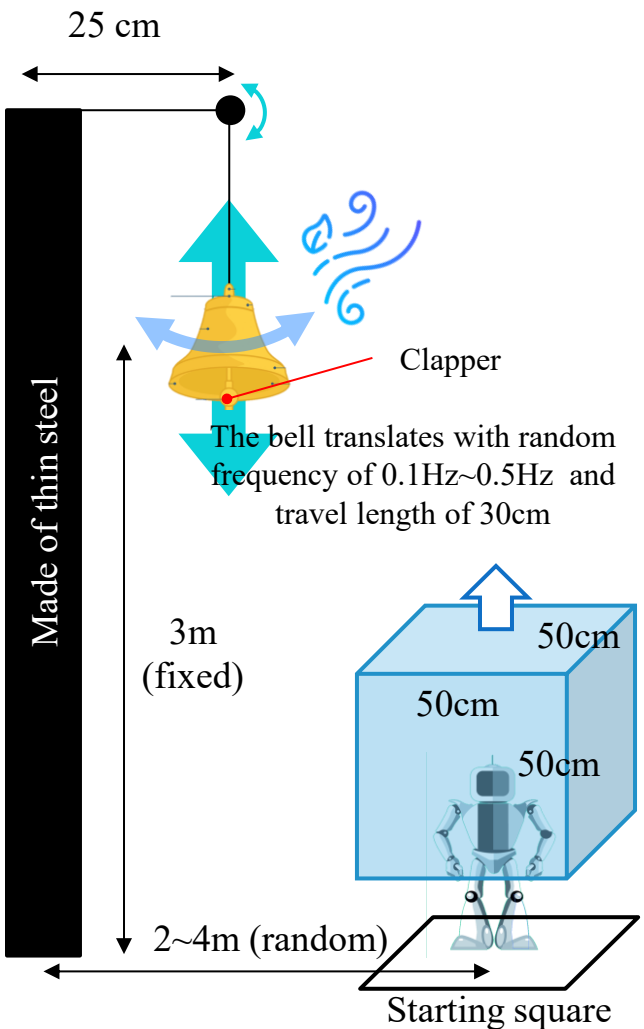
2026 Spring Mission

The challenge: Ring ma bell 2

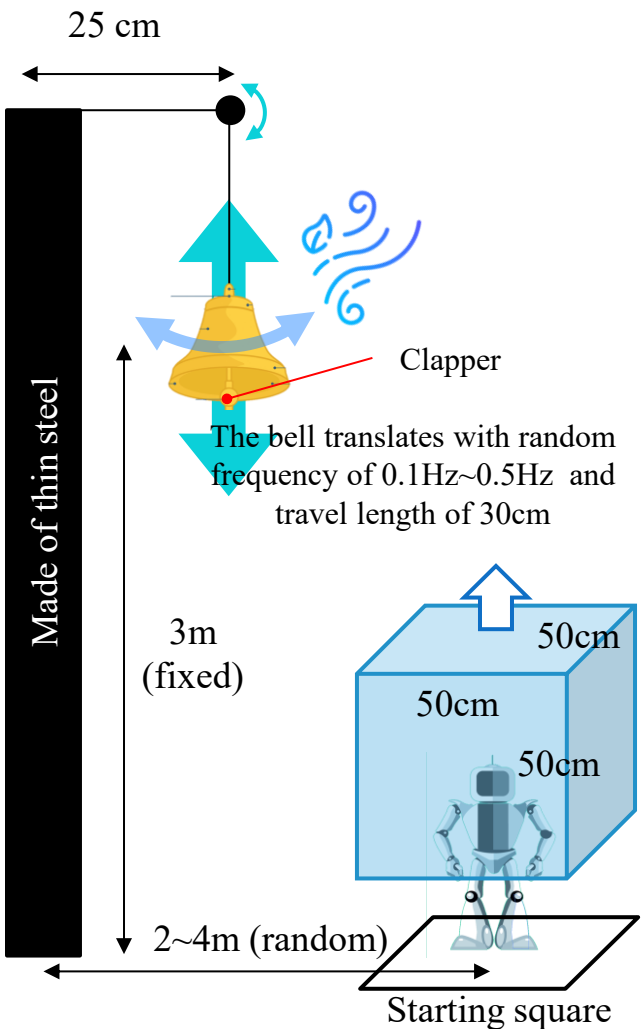


Challenge Guidelines

1. The goal is to ring the moving bell twice.
   - ✓ Definition of bell ringing: A component of the robot (which must be part of its measured weight) strikes the bell, causing the clapper to hit with the inner wall of the bell and produce an audible sound.
   - ✓ A minimum interval of 5 seconds is required between sounds caused by bell ringing.
2. Each team is allowed three trials of three minutes each. Teams failing to complete the task within these trials may be granted an additional opportunity.
3. A 10% deduction will be applied to the total score for each additional trial taken.
4. The order of the challenge is determined by drawing lots on the day of the event.
5. Weigh-ins are conducted immediately before each trial.
6. During their turn, each team may freely place their robot within the "starting square" designated by Referee 1.
7. Referee 1 will cover the robot with a 50x50x50 cm^3 box to verify its dimensions and determine eligibility.
8. Once Referee 1 removes the box, Referee 2 will start the timer. The robot must begin operation only after the box is removed. The trial ends after three minutes, as recorded by Referee 2.
9. Referee 3 records the time starting from the moment the robot begins operation. Do not touch the robot once the trial starts; any contact will lead to immediate disqualification.
10. If the robot does not operate at all, team members can adjust or reboot the robot during the trial. In that case, a 10-second penalty is added to the completion time of the trial.
11. If the goal is achieved within the time limit, the trial is recorded as a success, and the completion time is recorded by Referee 3 from the start until the moment the goal is achieved.

2026 Spring Mission

The challenge: Ring ma bell 2



Robot Manufacturing Guidelines

1. A development budget of KRW 2,000,000 is provided per group.
2. Only individual components may be purchased. The purchase of complete robot kits is strictly prohibited.
3. Any group exceeding the budget will automatically be ranked in last place. (A Bill of Materials (BOM) or detailed budget report must be included in the final submission.)
4. The robot must be self contained. (no external power, no external control)
5. The use of fuel, compressed gas, combustion, or other hazardous consumables is prohibited.
6. For any shooting robot, projectiles must be spherical and composed primarily of polymer, plastic, or rubber (e.g., tennis or ping-pong balls) to ensure safety.

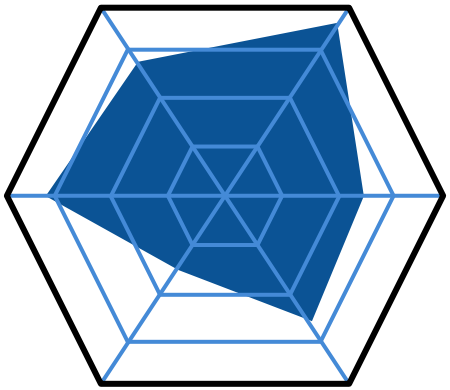
Evaluation

Grading Criteria of Final Demo (2026 spring)

Completion Time (25 pts):
Completion time is determined by the fastest successful trial among all attempts.
1st : 25 pts 2nd : 22 pts 3rd : 18 pts
4th : 15 pts 5th : 12 pts

Success Rate (25 pts):
Success rate is evaluated based on the number of times the task is achieved within a 3-minute time limit. Teams that fail to complete the task even once in three trials may attempt additional trials until they succeed, and if they complete the task during additional trials, it is recorded as 15 pts.
3/3 : 25 pts 2/3 : 20 pts 1/3 : 15 pts 0/3 : 10 pts

Size and weight (10+10 pts):
The robot must always remain within 50 * 50 * 50 cm^3 (10 pts). If the size is suspected to exceed the limit during the task, an investigation will be initiated. If it exceeds the limit, it will receive 0 points for size.
The robot's weight is measured before each trial, and the maximum recorded weight will be used for evaluation.
< 1kg : 10 (+5) pts < 2kg : 10 pts
< 5kg : 5 pts ≥ 5kg : 0 pts



Noise (10 pts):
The noise level is evaluated based on the committee's subjective assessment.

Creativity (10 pts):
The creativity is evaluated based on the committee's subjective assessment.

Safety (10 pts):
The safety is evaluated based on the committee's subjective assessment.
Unrecoverable after task completion (-5pts)